Master Upgrading PostgreSQL, In the Real World

redgate | About me

Pat Wright

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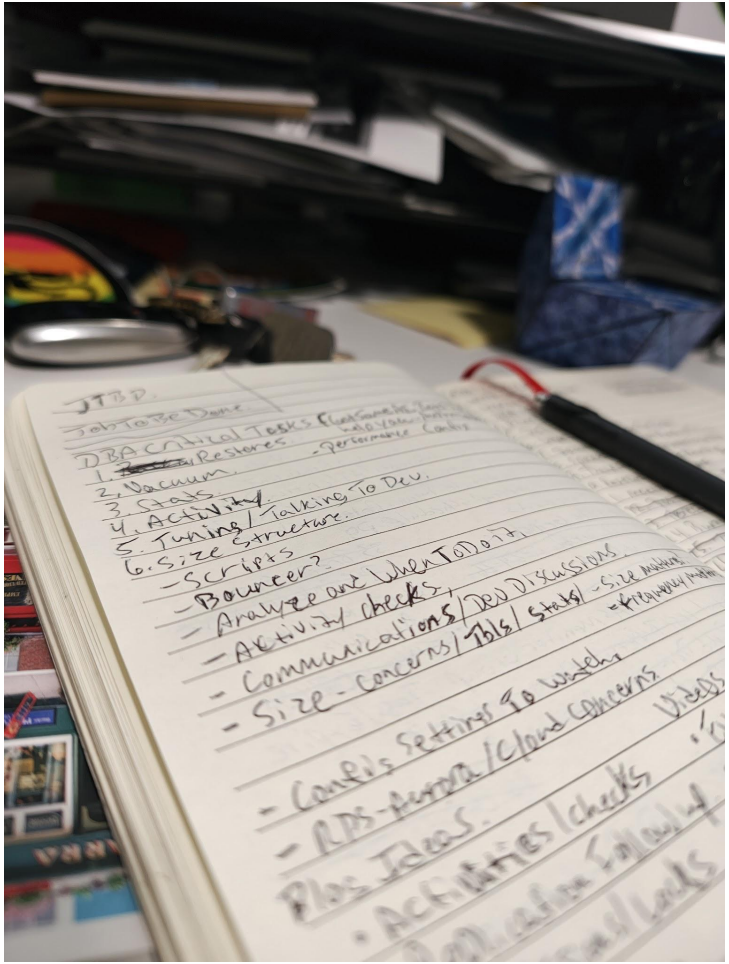
Master Upgrading PostgreSQL, Using Real World stories and examples

Agenda

- Why is this important?
- Lay the groundwork
- PG_Upgrade
- Key Tasks
- Large Scale Upgrades
- Real world fails
- Summary

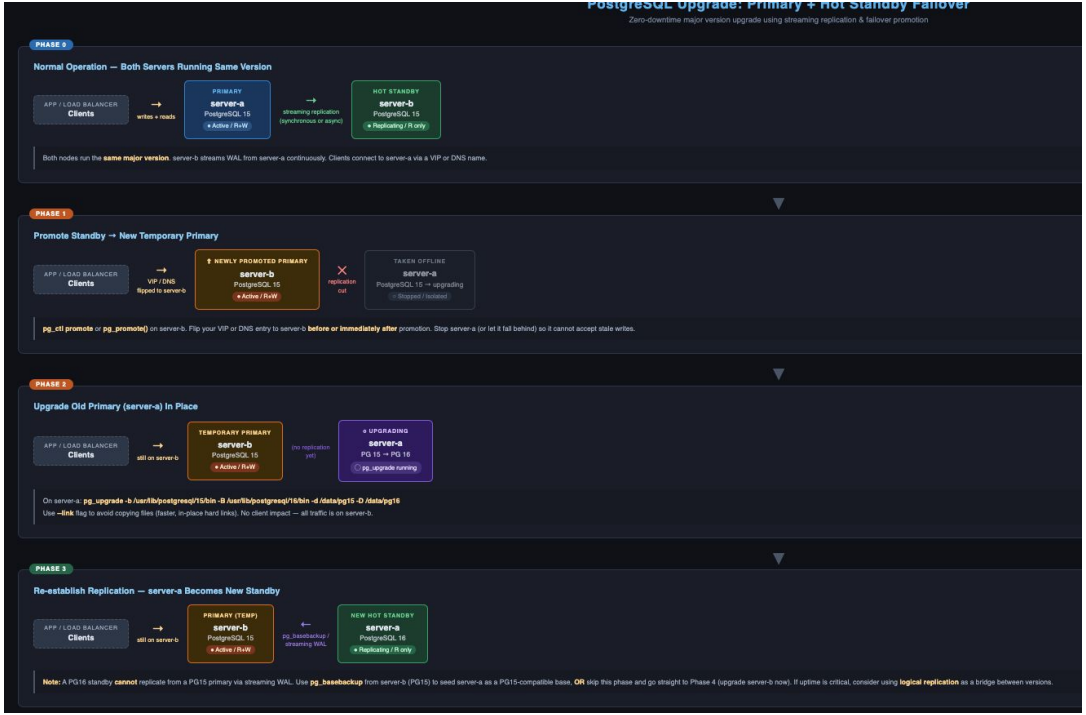


Why Upgrading matters?

- Critical for support
- Critical for features
- Critical for life of application
- It’s not easy.

PG_Upgrade

- Demo Process
- Stats still need to be updated
- Timings



Key’s for all Upgrade Projects

1. Find what is changing – Release Notes
2. Clone and Test – A lot , Seriously Lots
3. Dev upgrade and testing
4. Check with Teams/Communication
5. Determine Downtime
6. This is NOT the only upgrade

Running your Upgrade Incident

- EVERYONE involved
- Checklist/Task list.
- Go/No Go for Launch
- Point of no Return
- Automation!

LargeScale/Enterprise upgrades

- Cloud/Non Cloud
- Replication
- Failover/Switch
- Downtime
- Testing, Testing and more testing.
- Communication
- Automation

Failures

- Pg 15, “the AND issue”

```
pg_query(): Query Failed: ERROR: trailing junk after numeric literal at or near ...
```

After reviewing a sampling of the errors on this spreadsheet, I found that most of the screenshots reviewed show this error.

What happened is that in versions prior to Postgresql v15, this was an allowed action. However, in Postgresql v15, this was “fixed”. This error happens when a numeric literal or UUID is passed into the condition of a query and has another condition directly following (think use of AND/OR after passing in the condition value).

Here is a screenshot of the error.

```
HTTP_X_FORWARDED_FOR: 52.71.222.43, 172.70.135.160
REMOTE_ADDR: 10.81.214.197

Description:

pg_query(): Query failed: ERROR:  trailing junk after numeric literal at or near "200281A"
LINE 15:        c.id = 200281AND w.name Ilike '%Specials Nine%' OR w.i...
                     ^

Backtrace
-------------------
/var/www/Entrata/Eos/Connect/CDatabase.class.php::pg_query
```

This is taken from the Postgresql v15 release notes.

- Adjust JSON numeric literal processing to match the SQL/JSON-standard (Peter Eisentraut)

  This accepts numeric formats like `.1` and `1.`, and disallows trailing junk after numeric literals, like `1.type()`.

Failures

- The reporting query
- Rolling Forward with Production Code
- Weekend Warrior

Summary

- Planning is important.
- Get Used to Upgrading
- Built in Tools can be great!
- Find the unknowns to have less failures

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